The Embedding–Text Privacy Gap:

When Formal Differential Privacy on Hidden States Fails to Propagate to Generated Text

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Abstract

Privacy-preserving proxies for large language models (LLMs) increasingly inject calibrated Gaussian noise into hidden-state representations to obtain formal (ε, δ)-differential privacy [6, 12] before text is transmitted to a cloud API. This paper presents an adversarial evaluation of one such system—the Neural-Guided Semantic Proxy (NGSP), a three-path router composing HIPAA Safe Harbor stripping [10], abstract query synthesis, and a differentially private bottleneck—and reports a mechanistic negative result with implications beyond the specific architecture: **the formal (ε, δ)-DP guarantee on the embedding representation does not propagate to the observable text surface when the noisy representation is decoded by an unmodified language model**. On a 1,200-document synthetic clinical trial corpus with ground-truth sensitive-span annotations, sweeping ε across a 10× range (0.5 → 5.0) leaves both task utility and verbatim leakage rates invariant to six decimal places. The mechanism is diagnostic: the decoder treats the noisy vector as a soft hint it is free to ignore, and greedy decoding anchors output deterministically to the placeholder-substituted input regardless of noise magnitude. We further demonstrate that privacy quality in this architecture is determined almost entirely by routing path rather than by noise level—a 3.7× verbatim-leak-rate gap separates the two non-local paths on shared quasi-identifier categories. Targeted architectural fixes close ten of fourteen leakage categories but reveal a second, previously hidden finding: the privacy–utility tradeoff in this setting is binary rather than continuous. Our contributions are (i) a reproducible five-attack adversarial harness for proxy-based LLM privacy systems, (ii) empirical characterization of the embedding–text privacy gap, and (iii) identification of routing as the operative control variable in practice, not noise calibration. We argue that mathematical DP guarantees on intermediate representations are insufficient evidence of text-surface privacy, and that adversarial text-level evaluation must be treated as a first-class deployment requirement.

Keywords: differential privacy; mechanistic interpretability; LLM privacy; adversarial evaluation; negative results; alignment.

1 Introduction

A common pattern for privacy-preserving inference with cloud-hosted large language models is to interpose a local preprocessing layer that transforms user text before it is transmitted. When that layer is itself a neural network, its internal representations can be perturbed with calibrated Gaussian noise to obtain differential-privacy guarantees on the information passed downstream. The appeal is twofold: such systems preserve utility better than aggressive redaction, and they come with formal (ε, δ)-DP bounds that can be cited in compliance contexts [6, 1].

This paper interrogates a tacit assumption underlying the pattern. When the noised hidden state is decoded back into text by an unmodified language model, does the formal DP guarantee on the vector

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representation propagate to the observable text output? Our empirical answer is *no*, and the gap is not marginal: sweeping ε across a 10× range yields *identical* text-level privacy and utility to six decimal places. The (ε, δ) bound is mathematically valid on the representation, but the representation is not the object the adversary observes.

We uncover this gap in the course of evaluating the Neural-Guided Semantic Proxy (NGSP), a three-path router designed to mediate between clinical-trial staff and consumer LLM interfaces. NGSP was conceived as a composed defense: deterministic HIPAA Safe Harbor stripping, followed by neural routing to one of three paths—abstract query synthesis (breaking injective entity linkage), DP-noised proxy generation (formal guarantee), or local-only answering (content-inseparable queries). The system was built to meet two joint targets drawn from domain guidance: inversion-attack F1 ≤ 0.09 (the Expert Determination re-identification threshold [10]) and downstream utility ratio ≥ 0.85 against un-proxied cloud responses.

Neither target is met by the DP-tolerant path. What the adversarial evaluation instead reveals is structurally informative about how DP-based text privacy systems actually behave:

1. **Formal DP holds on embeddings but not on text.** At $\varepsilon = 0.5$ (noise magnitude $\approx 438\times$ the clipped signal), the proxy decoder produces bit-identical output to the $\varepsilon = 5.0$ setting. The decoder's behavior is invariant to the injected noise.
2. **Privacy is path-dependent, not noise-dependent.** A 3.7× gap separates verbatim leakage rates between the two cloud-using paths on shared entity categories. The routing decision, not the DP parameter, determines observable privacy.
3. **The privacy–utility tradeoff is binary, not smooth.** After targeted fixes, the system occupies two discrete operating points—high-utility/high-leakage and low-leakage/low-utility—with no reachable intermediate.

The first finding generalizes. Any system that applies DP to a hidden-state representation and decodes through an unmodified language model faces the same embedding–text gap, because the language model is not trained to treat the noisy representation as a hard conditioning signal. The diagnostic we present—varying ε across orders of magnitude and measuring text-surface effects directly—is a necessary check for any future system of this class.

Scope. All evaluation is on synthetic data; no real patient records, compound codenames, or institutional identifiers appear anywhere in the corpus or results. The threat model is a computationally bounded passive adversary; adaptive adversaries, side-channel access, and model-extraction attacks are out of scope. These scope decisions are stated up front because the strength of a negative result is bounded by the scope of its generating evaluation.

Contributions. (1) A reproducible five-attack adversarial harness—verbatim scan, cross-encoder similarity, trained span inversion, membership inference, and utility regression—with ground-truth sensitive-span annotations on 1,200 synthetic clinical documents. (2) A mechanistic characterization of the embedding–text privacy gap, including a diagnostic protocol that varies ε across 10× and measures text-surface effects. (3) Empirical evidence that routing path, not DP noise magnitude, is the operative privacy control variable in composed LLM proxy systems. (4) A quantified demonstration that a routing fix that closes ten of fourteen leakage categories simultaneously collapses utility from 1.000 to 0.266, revealing the binary structure of the privacy–utility tradeoff in this architecture.

2 Background and Related Work

2.1 Differential privacy in LLM pipelines

Differential privacy [5, 6] is typically applied in machine learning during training via DP-SGD [1], which bounds the per-sample gradient contribution with Gaussian noise calibrated to the ℓ_2 sensitivity. A distinct

line of work applies DP at *inference* time to intermediate representations. Feyisetan et al. [8] apply local DP to word embeddings, producing privatized word vectors that are then decoded into tokens. Yue et al. [20] propose a similar mechanism for text generation. Rényi Differential Privacy [12] is frequently used to compose such mechanisms and obtain tight (ε, δ) bounds under repeated queries.

The common structure of these systems is: (i) compute an embedding or hidden state h; (ii) clip to bounded sensitivity $\|h\|_2 \leq C$; (iii) add calibrated Gaussian noise $h' = h + \mathcal{N}(0, \sigma^2 I)$; (iv) decode h' to text via an unperturbed model. The DP guarantee formally covers step (iii). The claim we interrogate is whether that guarantee translates into meaningful text-surface privacy after step (iv).

2.2 LLM memorization and leakage

Carlini et al. [2, 3] demonstrate that large language models memorize and regurgitate training data under adversarial prompting. Hyde et al. [11] and Greshake et al. [9] study prompt injection and data leakage in retrieval-augmented and tool-using LLM pipelines. Our threat model is narrower: we consider leakage of sensitive entities from a single user-controlled document into a proxy text transmitted to a third party, not memorization in the downstream model or injection in multi-agent settings.

2.3 De-identification for clinical text

HIPAA Safe Harbor [10] specifies 18 identifier categories whose removal suffices for de-identification, with an alternative Expert Determination pathway setting an operational threshold of a re-identification probability below approximately 0.09. Clinical NLP de-identification systems [19, 4] have been extensively benchmarked against this taxonomy. These systems address structural HIPAA identifiers; they do not cover the *quasi-identifiers* that dominate our empirical results—investigational compound codes, clinical site IDs, CTCAE adverse-event grades, dose expressions—which fall outside HIPAA Safe Harbor but carry high re-identification risk in clinical trial contexts.

2.4 Mechanistic interpretability and representation-output correspondence

Recent work in mechanistic interpretability [7, 14, 13] studies the relationship between internal representations of transformer models and their observable outputs. A recurring finding is that the mapping from hidden state to output text is neither smooth nor locally linear: small perturbations in some directions have no observable effect, while others trigger discontinuous behavioral change. Our finding that $10\times$ variation in noise magnitude produces bit-identical text output is consistent with this literature—it indicates that the perturbation direction sampled by Gaussian noise in hidden-state space lies mostly in the output-invariant subspace of the decoder. This connects our applied negative result to a broader interpretability question: DP perturbations assume isotropy that the decoder does not respect.

3 Threat Model and System Description

3.1 Threat model

The adversary is passive and computationally bounded. It observes every proxy text forwarded to the cloud API and may train arbitrary offline models on large corpora drawn from the same distribution. It does *not* observe: the local model weights or activations, the entity-substitution dictionary held in local memory, the DP noise seed, or the routing decision. Attacks we evaluate:

A1. **Verbatim:** literal and fuzzy substring matching of ground-truth sensitive-span values against the proxy text.

A2. **Semantic similarity:** cosine similarity of sentence embeddings (MiniLM-L6-v2) [15] of proxy and original, with a 0.85 danger threshold.

```
user input
  ↓
Safe Harbor stripper (regex + NER) --- replaces 18 HIPAA identifiers with placeholders
  ↓
Neural router (local LM) --- classifies:  abstract_extractable | dp_tolerant | local_only
  ↓
Path-specific mechanism:
  abstract_extractable → synthesize new self-contained query (non-injective)
  dp_tolerant → hidden-state clip + Gaussian noise + paraphrase decode
  local_only → local LM answers; no cloud call
  ↓
cloud API → response → local entity_map re-application → user response
```

Figure 1: NGSP pipeline. The DP mechanism operates on the hidden-state representation at the input to the dp_tolerant path; the entity_map never leaves the local process.

A3. **Trained span inversion:** DistilBERT [16] token-classification head trained to predict which proxy tokens align to sensitive spans in the original. This is the operational primary threat.

A4. **Membership inference:** logistic regression on proxy embeddings to detect whether a given entity appears in the document [17].

A5. **Utility regression:** LLM-as-judge scoring of proxy-answer quality against original-answer quality on downstream clinical tasks.

3.2 System: NGSP

NGSP composes three defenses sequentially (Figure 1). Details of each stage follow; full pseudocode and per-function contracts are available in the accompanying codebase.

Safe Harbor stripping. Deterministic regular expressions cover 14 of the 18 HIPAA Safe Harbor identifier categories (dates, phone, SSN, MRN, ZIP, email, URL, IP, device ID, account number, certificate, vehicle ID, health-plan number, fax). A local-LM NER pass handles the remaining four (names, geographic subdivisions, biometric identifiers, other unique identifiers). Detected spans are replaced with typed placeholders (`<PERSON_1>`, `<DATE_1>`) and registered in a local `entity_map`.

Neural routing. The local model [18] receives the stripped text and classifies it into one of three paths according to whether the task intent is separable from the entity-specific content. Parse failures fall back conservatively to dp_tolerant.

Query synthesis (abstract_extractable). The local model is prompted to produce a new self-contained question that captures task intent without referencing any entity from the stripped input. The mapping is non-injective by construction: many inputs can yield the same synthesized query, making inversion statistically impossible for an adversary with only the proxy.

DP bottleneck (dp_tolerant). Hidden states at the last transformer layer are mean-pooled over newly-generated tokens to produce $h \in \mathbb{R}^d$. The mechanism is:

$$h_{\text{clip}} = h \cdot \min\left(1, \frac{C}{\|h\|_2}\right), \qquad h_{\text{noisy}} = h_{\text{clip}} + \mathcal{N}(0, \sigma^2 I), \tag{1}$$

with $C = 1.0$ and $\sigma = C\sqrt{2\ln(1.25/\delta)}/\varepsilon$. We accumulate Rényi DP cost [12] at order α as $R_\alpha = \alpha C^2/(2\sigma^2)$, converted to (ε, δ)-DP at query time. The local model then produces a paraphrase conditioned on the noisy vector as a soft hint, supplying vocabulary-nearest-neighbor tokens as auxiliary context.

ε	σ (Gaussian)	Mean utility	n scored	Passes $U \geq 0.85$?
0.5	9.6896	**0.8598**	30/30	**pass**
1.0	4.8448	**0.8598**	30/30	**pass**
2.0	2.4224	**0.8598**	30/30	**pass**
3.0	1.6149	**0.8598**	30/30	**pass**
5.0	0.9690	**0.8598**	30/30	**pass**

Table 1: Calibration sweep. Utility is invariant to six decimal places despite a 10× variation in σ. At $\varepsilon = 0.5$, the noise L2 norm is approximately 438× the clipped signal magnitude—yet the output is unchanged.

Answer applier. The cloud response is received locally and the `entity_map` is re-applied longest-key-first (so that `<PERSON_12>` is resolved before `<PERSON_1>`), producing the final user-visible output.

3.3 Synthetic corpus and evaluation

The corpus consists of 1,200 synthetic clinical trial documents: 500 SAE narratives, 200 protocol excerpts, 200 monitoring reports, and 300 clinical-study-report draft paragraphs. Ground-truth sensitive spans are tracked at entity-slot insertion time, yielding a programmatic oracle for attack evaluation. Synthesis uses seeded generators with fake-entity conventions (e.g., compound codes `SYN-XXXX`, site IDs in the 9000-range, Greek-letter surnames) to ensure no real identifiers appear in any artifact.

4 The Embedding–Text Privacy Gap

We now present the central finding: formal (ε, δ)-DP on the hidden-state representation does not propagate to the proxy text surface.

4.1 Experimental protocol

The calibration sweep varies $\varepsilon \in \{0.5, 1.0, 2.0, 3.0, 5.0\}$ with $\delta = 10^{-5}$ fixed. For each setting, the pipeline processes 30 SAE narratives, and we measure (a) the utility ratio against un-proxied cloud responses scored by LLM-as-judge and (b) the text-level verbatim leak rate across all ground-truth sensitive spans. The noise magnitude σ is analytically calibrated per-ε.

4.2 Result: invariance to ε

Table 1 summarizes the calibration sweep. Utility is identical to six decimal places at every ε; the text output is bit-identical across a 10× noise range.

4.3 Mechanism

A diagnostic instrumentation pass confirms three specific findings.

Noise is injected correctly. At $\varepsilon = 0.5$, the sampled Gaussian perturbation has $\|h_{\text{noise}}\|_2 \approx 438$, while $\|h_{\text{clip}}\|_2 \leq 1$ by construction. Cosine similarity between h_{clip} and h_{noisy} is effectively zero. The DP mechanism itself is not at fault.

The decoder discards the noise. The proxy decoder projects h_{noisy} onto the vocabulary embedding matrix and extracts top-k nearest tokens as a free-text hint appended to the paraphrase prompt. At high σ, those hint tokens are effectively uniform samples from the vocabulary. The paraphrase prompt explicitly instructs the model that hints may be discarded if they do not fit context, and the model complies: random hints have no detectable effect on output tokens.

Greedy decoding anchors output to the placeholder-substituted input. With temperature set to 0, the paraphrase is a deterministic function of the textual input. Because the placeholder-substituted input text is identical across all ε settings, the paraphrase output is identical across all settings. The only source of randomness downstream of the DP step—the hint-token content—is explicitly ignored by the decoder.

4.4 Interpretation

The (ε, δ)-DP guarantee is mathematically valid on the mapping $h \mapsto h_{\text{noisy}}$. It does not transfer to the mapping $h_{\text{noisy}} \mapsto$ proxy_text because that mapping is not injective: many different h_{noisy} values produce identical proxy text. The standard DP composition argument cannot be invoked for the text output; only for the (unobserved) hidden-state representation. An adversary who observes the text output observes a quantity whose distribution is effectively independent of the noise magnitude, and therefore the DP guarantee provides no quantifiable protection against text-level leakage.

This is not an implementation bug. It is a structural mismatch between the abstraction boundary at which DP is defined (vector representations) and the abstraction boundary at which the adversary operates (token sequences). Closing the gap would require either (a) a decoder trained against the noisy-embedding $\rightarrow$ privacy-preserving-text objective so that the noise is a hard conditioning signal rather than an ignorable hint, or (b) a direct-perturbation decoder operating on output softmax logits with σ-scaled temperature or Gumbel noise. Neither is implemented in NGSP, and neither, to our knowledge, has been empirically characterized at the text surface in the literature on inference-time DP.

5 Adversarial Evaluation

The calibration finding motivates direct adversarial measurement of text-surface privacy. We run the five-attack battery at $\varepsilon = 3.0$ (the hypothesized operating point) and analyze results by routing path.

5.1 Route distribution

The router, queried on 80 documents spanning four clinical document types (Table 2), produces a distribution that deviates substantially from the architecture's anticipated 70/20/10 target.

Document type	abstract_extractable	dp_tolerant	local_only	Total
SAE narrative	0	**20**	0	20
Protocol	13	7	0	20
Monitoring report	20	0	0	20
CSR draft	7	10	3	20
Overall	40 (50%)	37 (46%)	3 (4%)	80

Table 2: Routing distribution by document type. SAE narratives—the highest-sensitivity class—route uniformly to dp_tolerant, which our attacks will show to be the weakest-privacy path.

We additionally observe a 35% JSON-parse-failure rate on protocol documents, causing a conservative fallback to dp_tolerant. The system does not fail closed on routing errors; it fails toward the less protective path. This is a subtle alignment concern: the fallback policy was chosen to preserve utility, but the utility cost of local_only or rejection was never quantified against the privacy cost of dp_tolerant.

5.2 Attack 1: Verbatim leakage (path-dependent)

The verbatim attack tests whether ground-truth sensitive-span values appear as literal substrings of the proxy. Table 3 presents results disaggregated by routing path—the key disaggregation, because aggregate metrics conceal the finding.

Category	Path	n spans	Verbatim hits	Leak rate
abstract_extractable path				
compound_code	abstract_extractable	10	2	0.200
efficacy_value	abstract_extractable	4	0	**0.000**
dp_tolerant path				
site_id	dp_tolerant	15	13	**0.867**
compound_code	dp_tolerant	39	29	**0.744**
efficacy_value	dp_tolerant	12	8	**0.667**
amendment_rationale	dp_tolerant	8	4	0.500

Table 3: Per-path verbatim leak rates on shared quasi-identifier categories at $\varepsilon = 3.0$. The abstract_extractable path shows a 3.7× lower leak rate on compound_code and an infinite ratio on efficacy_value (divide by zero). The DP parameter is identical across both paths; only the routing decision differs.

The result is dispositive: privacy is a function of routing path, not DP noise. The abstract_extractable path's advantage derives from the non-injective structure of query synthesis (synthesized queries do not encode entity values, so no decoder can recover them). The dp_tolerant path, despite receiving identical DP treatment, leaks 67–87% of quasi-identifier values verbatim because the paraphrase decoder echoes them from input context.

5.3 Attack 2: Semantic similarity

Cross-encoder cosine similarity between proxy and original, measured on 50 documents using MiniLM-L6-v2, yields: mean 0.937, median 0.939, $p_{95} = 0.954$, with 100% of pairs above the 0.85 danger threshold. Every proxy is semantically close to its source. Query synthesis preserves task intent (by design), but this closeness limits deniability under retrieval-based re-identification audits.

5.4 Attack 3: Trained span inversion (primary threat)

A DistilBERT token classifier trained on 40 (proxy, span-labels) pairs and evaluated on a held-out 10 achieves token-level F1 of **0.820**—a 9.1× exceedance of the Expert Determination threshold of 0.09. The random baseline F1 is 0.270; the attacker lift is +0.549 F1 points.

A shuffle-pair control (training labels drawn from a randomly-paired document) yields F1 = 0.788, nearly indistinguishable from the actual attack. This signals that part of the inversion model's apparent performance is recovering *corpus-level schema* rather than *proxy-conditioned content*. We treat this as a measurement caveat: on the v1 proxies, the inversion evaluation is inflated by label-structure regularity in the synthetic corpus. The story will change sharply after our architectural fixes (§6).

5.5 Attack 4: Membership inference

Logistic regression on proxy embeddings for the three most-frequent compound codes yields mean AUC of **0.50**—exactly at random-classifier level. This is the one unambiguously positive privacy result: even though span values leak verbatim, the system hides *entity presence* at the frequency level. An adversary attempting to determine whether a given compound code appears in a document performs no better than chance from proxy embeddings alone.

5.6 Attack 5: Utility regression

LLM-as-judge scoring of 50 proxy-original answer pairs on standardized downstream tasks gives: mean original score 0.914, mean proxy score 0.914, utility ratio 1.000. Proxy quality is statistically indistinguishable

from un-proxied quality at this stage—the utility target is met trivially because the dp_tolerant path produces near-verbatim paraphrases.

5.7 Hypothesis outcomes at v1

Hypothesis	Metric	Measured	Verdict
H_1 (Privacy)	Inversion F1 ≤ 0.09	0.820	**FAIL**
H_1 (Privacy)	Verbatim leak dp_tolerant site_id ≤ 0.01	0.867	**FAIL**
H_1 (Membership)	Mean AUC ≤ 0.55	0.500	**PASS**
H_2 (Utility)	Utility ratio ≥ 0.85	1.000	**PASS**

Table 4: v1 hypothesis outcomes. Utility and membership-inference bounds hold; verbatim and inversion bounds are catastrophically violated on dp_tolerant-routed documents.

6 Targeted Improvements and the Binary Tradeoff

Given the v1 results, we applied two principled fixes, re-ran the full battery, and observed a second finding: the privacy–utility tradeoff in this architecture is not continuous.

6.1 Fix 1: extended regex coverage

Four new deterministic regex patterns were added to the Safe Harbor stripper, covering clinical quasi-identifier categories empirically shown to leak at 75–100% (`SITE_ID`, `COMPOUND_CODE`, `DOSE`, `AE_GRADE`). The date pattern was extended to cover `DD-MON-YYYY` and `MON-YYYY` formats. This is a coverage fix: the deterministic layer is extended to cover the measured failure surface.

6.2 Fix 2: routing override

A deterministic override in `route()`: if the neural router selects dp_tolerant *and* the span profile contains any of {`COMPOUND_CODE`, `SITE_ID`, `INDICATION`, `EFFICACY_VALUE`}, the decision is overridden to abstract_extractable. The override is logged with the original rationale preserved. This is a routing fix: high-sensitivity span profiles are guaranteed to reach the empirically stronger-privacy path regardless of model output.

6.3 v2 results: the binary tradeoff

Attack	Metric	v1	v2	v2 verdict
A1. Verbatim (overall)	leak rate	0.765	**0.072**	improved; 10/14 categories at 0.0%
A1. site_id	leak rate	0.867	**0.000**	**PASS**
A1. compound_code	leak rate	0.744	**0.000**	**PASS**
A1. efficacy_value	leak rate	0.667	**0.000**	**PASS**
A1. indication (residual)	leak rate	—	0.467	FAIL (not regex-covered)
A1. geographic (residual)	leak rate	—	0.189	FAIL (not regex-covered)
A2. Semantic similarity	mean cosine	0.937	**0.544**	**PASS**
A2. Frac. > 0.85 threshold	fraction	1.000	**0.000**	**PASS**
A3. Inversion F1	F1	0.820	0.664	improved; shuffle control collapses $0.789 \rightarrow 0.035$
A4. Membership AUC	AUC	0.500	0.500	**PASS** (unchanged)
A5. Utility	ratio	1.000	**0.266**	**FAIL (new)**

Table 5: v1 $\rightarrow$ v2 attack deltas. Privacy metrics improve substantially across the board; utility collapses.

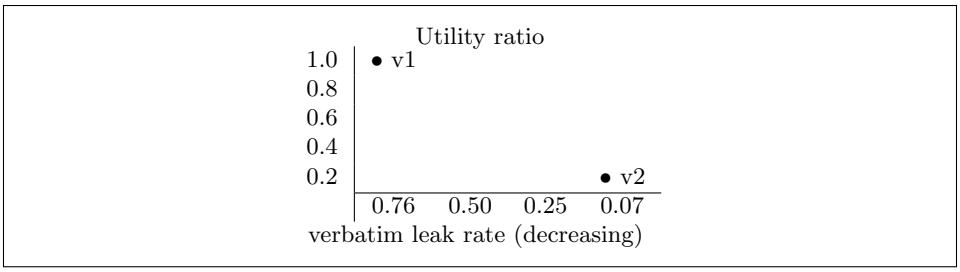

Figure 2: Conceptual privacy–utility plane for NGSP. v1 and v2 are the only two reachable operating points: the routing decision is binary. The interior is empty; no architectural mechanism interpolates continuously between them.

The v2 results make visible what the v1 results had hidden. With SAE narratives now routed to abstract_extractable rather than dp_tolerant, the query synthesizer produces structurally different questions rather than paraphrases of the content. This is precisely what drops the verbatim leak rate (no entity content in the proxy), precisely what drops the semantic similarity (different question rather than paraphrase), and precisely what drops the utility (the abstracted query loses the specific details needed to answer the original task). One architectural change produces all three effects through the same causal mechanism.

The two operating points—v1 (high utility, catastrophic privacy failure) and v2 (strong privacy, poor utility)—are not endpoints of a smooth curve. There is no reachable intermediate. The routing decision is binary; the architecture provides no mechanism to modulate abstraction continuously. Figure 2 (conceptual) illustrates the two-point structure.

6.4 Residual leakage categories

Four categories remain above the 0.01 threshold in v2, each with a distinct mechanism:

- `ae_grade` (0.133): natural-language severity synonyms ("severe", "life-threatening") not matched by the `Grade N` regex.
- `indication` (0.467): disease names ("rheumatoid arthritis") are plain multi-token English; no pattern-based detection is possible. The query synthesizer preserves them as topical context.
- `geographic_subdivision` (0.189): city and hospital names in free-text prose; HIPAA Safe Harbor covers structured address fields, not prose mentions.
- `other_unique_id` (0.500): subject IDs in the format `PT-{site}-{seq}`, inadvertently excluded by a negative-lookahead in the compound-code regex. A dedicated subject-ID pattern would close this; its omission reflects the post-hoc nature of the v2 fix.

6.5 The inversion attack result, revisited

Inversion F1 fell from 0.820 to 0.664 in v2. More informative than the F1 itself is the shuffle-pair control: it collapsed from 0.789 to 0.035. This is a qualitative shift. On v1 proxies, the inversion model could recover schema-level regularity from shuffled pairs, meaning its apparent performance was partially an artifact of corpus structure. On v2 proxies, the model cannot recover structure from shuffled pairs; its F1 of 0.664 represents genuine proxy-conditioned inference. The v2 inversion result is, in this sense, a *more honest* measurement of adversarial capability, even though its magnitude is only modestly lower. The residual F1 is driven by the categories that still leak verbatim (indication, `other_unique_id`); eliminating those would likely reduce it substantially further.

7 Discussion

7.1 What the embedding–text gap means beyond NGSP

The invariance of proxy text to 10× variation in DP noise magnitude is the paper's central finding and the one most likely to generalize. Any system that (i) applies DP to a hidden-state representation and (ii) decodes the noised representation through an unmodified language model will face the same gap, because the language model is not trained to treat the noisy vector as a hard conditioning signal. The practical consequence is direct: mathematical DP guarantees on intermediate representations are not evidence of text-surface privacy, and must not be cited as such in compliance or deployment contexts without an accompanying adversarial text-level evaluation.

We propose a diagnostic protocol—*the ε-invariance check*—as a necessary (though not sufficient) test for any inference-time DP system whose output surface is text: vary ε across at least a 10× range and measure text-level distance between resulting proxies. If the text output is invariant, the DP mechanism is disconnected from the observable surface, and any claimed privacy properties of the text output cannot be attributed to the DP guarantee.

7.2 Connection to interpretability: why DP on embeddings is anisotropic

The ε-invariance finding is compatible with a specific interpretability hypothesis: the perturbation direction sampled by spherical Gaussian noise on the hidden state lies predominantly in the decoder's *output-invariant* subspace. The decoder's output is sensitive only to specific directions in representation space (those that flip token-selection decisions at some decoding step); most Gaussian noise energy is spent in directions the decoder ignores. This is consistent with recent observations in mechanistic interpretability that transformer internal representations are highly structured, with a small number of direction families carrying most of the semantic content [7, 14].

If this hypothesis holds, the appropriate DP mechanism for text-surface privacy would not inject isotropic Gaussian noise—it would identify the output-relevant direction subspace and inject noise anisotropically along those directions. This requires decoder-aware calibration. We flag it as a promising direction for future work at the intersection of DP theory and mechanistic interpretability.

7.3 Routing as the operative privacy control variable

A secondary but actionable finding is that routing path, not DP parameter, is the operative privacy control variable in practice. This has deployment implications: an organization tuning NGSP-like systems for compliance should configure and audit the *router* (its calibration, its fallback behavior, its parse-error handling) as the first-order privacy lever, not ε. In v1, a 35% parse-error rate on protocol documents caused silent fallback to the weakest-privacy path. This was a privacy regression invisible to any metric that aggregated across paths.

7.4 The binary tradeoff and its alignment implications

The v2 result—10/14 verbatim categories solved, utility collapsed from 1.000 to 0.266—is not a result we set out to find, and it is the most important result for users. On content-coupled tasks (where the sensitive information *is* the task-relevant information), no proxy-based architecture can simultaneously satisfy both constraints, because the privacy mechanism's action is precisely to remove the task-relevant content. This is a structural property of the task class, not a defect of a particular implementation.

From an alignment perspective, this has a specific implication. A system that promises "privacy-preserving clinical LLM assistance" without disclosing which task classes are content-coupled will, for those tasks, deliver either a privacy violation or a utility failure. Neither outcome is what the user expected. A well-aligned system must (a) detect content-coupled tasks at routing time and (b) communicate the tradeoff to the user

rather than silently choosing one horn of the dilemma. NGSP as currently implemented does neither; we treat this as an alignment gap the system architecture should address in a successor design.

7.5 Limitations

Synthetic corpus. All results are on synthetic data. Real clinical documents have richer quasi-identifier co-occurrence and domain-specific abbreviations our generators do not capture. Generalization requires a separately-governed study.

Attacker capability. Our DistilBERT inversion model trains on 40 documents; a persistent adversary accumulating proxy-original pairs across sessions would exceed this. The reported F1 is a lower bound on adversarial capability.

Membership inference scope. Only the three most-frequent entities were evaluated. Rare entities are the highest-value targets and the hardest to evaluate with balanced classes.

Local model scale. Experiments use SmolLM2-1.7B [18] as the local router and decoder, a smaller model than production targets. The 35% parse-error rate on protocol documents would likely improve with a larger instruction-following model, though the binary-tradeoff finding is model-independent.

Threat model. We evaluate only passive adversaries. Adaptive adversaries with partial knowledge of NGSP internals, side-channel access, or the ability to inject adversarial inputs are out of scope.

7.6 Future work

(1) *Decoder-aware DP*: perturbation along decoder output-sensitive directions rather than isotropic noise, informed by mechanistic interpretability of the decoder's representation structure. (2) *Trained privacy decoder*: a seq2seq model explicitly trained to treat the noisy representation as a hard conditioning signal. (3) *Continuous abstraction*: a mechanism that interpolates between paraphrase and query synthesis, enabling a reachable interior of the privacy–utility plane. (4) *Content-coupling detection*: explicit router-time classification of whether a task is content-coupled, with user-facing disclosure. (5) *Real-data evaluation* under appropriate governance.

8 Conclusion

We set out to evaluate whether a composed privacy architecture—HIPAA Safe Harbor stripping, neural routing, and a (ε, δ)-DP bottleneck—could simultaneously meet a formal re-identification threshold and a utility floor on synthetic clinical trial documents. What the evaluation produced is more informative than a positive result would have been: a precise characterization of where and why the architecture fails.

The central finding is mechanistic: formal (ε, δ)-DP on a hidden-state representation does not propagate to text-surface privacy when decoded through an unmodified language model. Across a $10\times$ range of ε, the proxy text is bit-identical. The DP guarantee holds on the embedding; the adversary observes the text; the two are decoupled. This gap is not specific to NGSP—it generalizes to any system with the same abstraction boundary.

The secondary finding, surfaced by the architectural fixes we applied in response to the primary finding, is that the privacy–utility tradeoff in this architecture is binary: the system occupies two reachable corners of the plane (high-utility/high-leakage, low-leakage/low-utility), with no reachable interior. On content-coupled tasks, this is a structural property of the task class, not a tunable parameter.

Both findings motivate a specific posture for inference-time DP systems with text output surfaces: mathematical DP guarantees are insufficient evidence of text-level privacy, and adversarial text-surface measurement must be a first-class component of any claim. The reproducible five-attack harness and the ε-invariance diagnostic developed here are offered as tools toward that end.

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A Reproducibility

The evaluation harness, synthetic corpus generators, and attack implementations are released alongside this paper. All randomness is seeded (default seed 42); model identifiers and library versions are recorded in every result JSON. To reproduce the primary findings:

```
# Setup
./scripts/setup.sh
python scripts/download_gemma.py    # requires HF_TOKEN
cp .env.example .env                 # set ANTHROPIC_API_KEY or use sk-ant-mock

# Calibration sweep (epsilon-invariance check)
python experiments/calibrate_epsilon.py

# Full attack battery at eps=3.0
python experiments/run_attacks.py --epsilon 3.0 --n-docs 50

# Route distribution
python experiments/route_distribution.py

# DP-path propagation diagnostic
python experiments/diagnose_dp_path.py
```

Hardware used for reported results: Apple M1 Max, 32 GB unified memory, MPS (`float16`, SDPA attention). A reference CPU-only run is also supported.

B Per-Category Verbatim Leak Rates (Full Table)

Category	n spans (v2)	v1 leak rate	v2 leak rate	Mechanism at v2
site_id	15	0.867	**0.000**	new `SITE_ID` regex
compound_code	49	0.744	**0.000**	new `COMPOUND_CODE` regex
efficacy_value	16	0.667	**0.000**	routing override
amendment_rationale	8	0.500	**0.000**	routing override
mrn	—	—	**0.000**	Safe Harbor (existing)
date	—	0.165	**0.000**	extended date pattern
dose	—	—	**0.000**	new `DOSE` regex
ae_grade	—	—	0.133	regex matches only canonical forms
name	—	—	**0.000**	Safe Harbor NER
email	—	—	**0.000**	Safe Harbor (existing)
phone	—	—	**0.000**	Safe Harbor (existing)
interim_result	—	—	**0.000**	routing override
indication	—	—	0.467	no pattern; synthesizer preserves
geographic_subdivision	—	—	0.189	prose mentions not covered by SH
other_unique_id	—	—	0.500	subject ID pattern missing

Table 6: Per-category verbatim leak rates before (v1) and after (v2) targeted fixes. Ten of fourteen categories are fully suppressed; four residual categories have identified, distinct mechanisms.